

Introduction to Tuples in Python

Prepared by:
Asif Farooq

Introduction to Tuples

- ▶ Tuples are a collection of ordered, immutable elements.
- ▶ They can store multiple types of data (integers, strings, etc.).
- ▶ Tuples are defined using parentheses ().

Key Characteristics of Tuples

- ▶ Ordered: Items have a defined order, and that order will not change.
- ▶ Immutable: Once created, elements cannot be changed (no adding, removing, or modifying items).
- ▶ Allow Duplicates: Tuples can contain duplicate values.
- ▶ Indexing: Like lists, items in tuples are indexed starting from 0.

Creating a Tuple

- ▶ Syntax:

```
my_tuple = (1, 2, 3,3)
print(len(tuple))
```

- ▶ Empty Tuple:

```
empty_tuple = ()
```

- ▶ Single Element Tuple:

```
single_element_tuple = (5,)
```

- ▶ Using tuple constructor

```
thistuple = tuple(("apple", "banana", "cherry"))
print(thistuple)
```

Accessing Tuple Elements

Using Indexing:

```
my_tuple = (10, 20, 30)  
print(my_tuple[0]) # Output: 10
```

Negative Indexing:

```
print(my_tuple[-1]) # Output: 30  
print(mytuple[-4:-1])  
del mytuple
```

Tuple Slicing

► Slicing a Tuple:

```
my_tuple = (10, 20, 30, 40, 50)
```

```
sliced_tuple = my_tuple[1:4]
```

```
print(sliced_tuple)
```

Tuple Operations

- ▶ Concatenation:

```
tuple1 = (1, 2)
```

```
tuple2 = (3, 4)
```

```
result = tuple1 + tuple2
```

- ▶ Repetition:

```
my_tuple = (1, 2)
```

```
result = my_tuple * 3
```

Packing and Unpacking of Tuples

- ▶ *When we create a tuple, we normally assign values to it. This is called "packing" a tuple*

```
fruits = ("apple", "banana", "cherry")
```

- ▶ *When we extract the values back into variables from a tuple it is known as "unpacking":*

```
(green, yellow, red) = fruits  
print(green)  
print(yellow)  
print(red)
```

- ▶ Using * operator

```
fruits = ("apple", "banana", "cherry", "strawberry", "raspberry")
```

```
(green, yellow, *red) = fruits
```


Tuple Methods

- ▶ `Count()`: Returns the number of occurrences of an element.

```
my_tuple = (1, 2, 2, 3)
print(my_tuple.count(2))
```

- ▶ `Index()`: Returns the index of the first occurrence of an element.

```
my_tuple = (10, 20, 30)
print(my_tuple.index(20))
```

Why Use Tuples?

- ▶ Advantages:
 - ▶ Immutable, which means they are safe from accidental modification.
 - ▶ Faster than lists for read-only operations.
 - ▶ Useful for data integrity when the collection shouldn't change.

Comparing Tuples and Lists

- ▶ Similarities:

- Both store multiple elements.

- Both allow indexing and slicing.

- ▶ Differences:

- Lists are mutable; tuples are immutable.

- Lists use square brackets [], while tuples use parentheses ().

Programs on List & Tuples

- ▶ Write a program to store the name of subjects in a list
- ▶ Write a program to accept marks of a student and calculate their sum
- ▶ Write a program to sort the names of fruits
- ▶ Write a program to count the number of fruits in a tuple

THANK YOU